Category	Item	Responsible
Coordination	GC to coordinate with EC, CBRE & IOT onboarding call to review start-up process and site readiness	GC
Conduit	Wall Boxes - conduit stub-up with pull string/line (for all lighting devices and all HVAC devices) see E and M pages for details	EC
Conduit	Conduit stup up through roof curb for control wire(s) with pull string/line.	EC
Mounting	LCP Panel(s) and override(s).	EC
Power	LCP Panel(s) and override(s).	EC
Mounting	Shades Panel(s) and Override(s)	EC
Power	Shades Panel(s) and Override(s)	EC
Power	BMS control panel(s).	EC
Coordination	GC to coordinate with MC, CBRE & IOT onboarding call to review start-up process and site readiness	GC
Wiring	Thermostat wiring to HVAC system(s). Cable sized to include 2 spare conductors and leave 1 pull wire for BMS network.	MC
Mounting	POL-224 economizer controller to be fully installed and configured with accessories per IOM. Include enthalpy OA & RA sensors	MC
Mounting	Temporary thermostat control for AHU's & RTU's at the space	MC
Wiring	CO2 sensor cabling for RTU DCV control.	MC
Wiring	CO2 sensor cabling for SSAHU Controller.	MC
Start-up & Documentation	Manufacturer specified start-up and report of HVAC systems.	MC
Wiring	Install and configure 24V interface for minisplit based on manufacturer	MC
Start-up & Documentation	Submit copy or picture of start-up report to CI.	MC
Wiring	BMS and BACnet network control system wiring.	CI
Fire, Life Safety	Duct Smoke detector - See note 1.	MC
Fire, Life Safety	Fire alarm shutdown relay/connection to FACP - See note 2.	FAC
Mounting	Relief air damper and actuator.	MC
Mounting	Relief air fan or return air fan.	MC
Mounting	Outside air damper and actuator.	MC
Start-up & Documentation	As-built drawings to ME	MC
Mounting	Replace temporary thermostats with BMS thermostats (return temp to MC).	CI
Start-up & Documentation	BMS network start-up & LCP system commissioning	EC/CI
Start-up & Documentation	BMS controls & HVAC system commissioning	MC/CI
Warranty	HVAC Equipment on-board controls/sensors labor warranty.	MC
Warranty	Controls integration and user interface warranty & tech support.	CI
Warranty	HVAC equipment onboard controls/sensors parts warranty & tech support.	MANUFACTURER

MATRIX NOTES:

1. WHERE A DUCT SMOKE DETECTOR IS SHOWN, THE FOLLOWING ARE THE RESPONSIBILITIES:
   - **A.** WHERE THE BUILDING HAS A FIRE ALARM SYSTEM, THE DUCT SMOKE DETECTOR SHALL BE PROVIDED BY THE ELECTRICAL CONTRACTOR, INSTALLED BY THE MECHANICAL CONTRACTOR, AND WIRED/CONNECTED BY THE FIRE ALARM CONTRACTOR.
   - **B.** WHERE THE BUILDING DOES NOT HAVE A FIRE ALARM SYSTEM, THE DUCT SMOKE DETECTOR SHALL BE PROVIDED BY THE MECHANICAL CONTRACTOR, INSTALLED BY THE MECHANICAL CONTRACTOR, AND WIRED/CONNECTED BY THE MECHANICAL CONTRACTOR.

2. WHERE A CONTROL MONITOR RELAY IS SHOWN FOR AN HVAC UNIT SHUTDOWN, THE FOLLOWING ARE THE RESPONSIBILITIES:
   - **A.** WHERE THE BUILDING HAS A FIRE ALARM SYSTEM, THE CONTROL MONITOR RELAY SHALL BE PROVIDED BY THE ELECTRICAL CONTRACTOR, INSTALLED BY THE FIRE ALARM CONTRACTOR, AND WIRED/CONNECTED BY THE FIRE ALARM CONTRACTOR.
   - **B.** WHERE THE BUILDING DOES NOT HAVE A FIRE ALARM SYSTEM, THE CONTROL MONITOR RELAY SHALL BE PROVIDED BY THE MECHANICAL CONTRACTOR, INSTALLED BY THE MECHANICAL CONTRACTOR, INSTALLED BY THE MECHANICAL CONTRACTOR, AND WIRED/CONNECTED BY THE MECHANICAL CONTRACTOR.

IOT BMS NOTE:

IOT SHALL PROVIDE BMS. COORDINATE WITH IOT FOR SUPPORT PRICING. CONTACT JILL MASON AT (512)-649-5142.

MECHANICAL CONTRACTOR TO SCHEDULE ON BOARDING SESSIONS WITH IOT PRIOR TO INSTALL.

1 M-503 IOT MATRIX
NOT TO SCALE

LEGEND:
CI: CONTROLS INSTALLER
EC: ELECTRICAL CONTRACTOR
MC: MECHANICAL CONTRACTOR
ME: MECHANICAL ENGINEER
GC: GENERAL CONTRACTOR
FAC: FIRE ALARM CONTRACTOR
OEM: ORIGINAL EQUIP MANUFACTURER

GENERAL NOTES

1. THE "RESPONSIBLE" COULMN BEARS THE RESPECTIVE CONTRACTOR TO PROVIDE AND INSTALL UNLESS OTHERWISE NOTED.
2. WHERE THERE IS A CONFLICT BETWEEN THE SPECIFICATIONS AND THIS MATRIX, THE MATRIX SHALL TAKE PRECEDENCE.
3. ALL CONDUIT SHALL BE INSTALLED PER NEC AND SPECIFICATIONS. ALL STUB UPS SHALL BE STUBBED UP INTO AN ACCESSIBLE CEILING. THERE SHALL BE NO CONCEALED JUNCTION BOXES

OCCUPIED MODE:
THE WALL MOUNTED PROGRAMMABLE THERMOSTAT SHALL START THE SUPPLY FAN, SUBJECT TO INTERNAL AC UNIT SAFETIES AND SMOKE DETECTOR INTERLOCK (WHERE REQUIRED). THE SPACE TEMPERATURE SENSOR SHALL CYCLE ON THE COMPRESSOR(S) TO MAINTAIN COOLING SETPOINT (74° F - ADJUSTABLE). UPON A DROP IN SPACE TEMPERATURE DURING WINTER MONTHS, THE UNIT SHALL STAGE ON THE COMPRESSOR(S) TO MAINTAIN SPACE TEMPERATURE SETPOINT (70°F - ADJUSTABLE). IF THE HEAT PUMP CANNOT SATISFY SPACE TEMPERATURE SETPOINT, THE ELECTRIC HEAT SHALL STAGE ON. DURING OCCUPIED HOURS, IF THE CO2 LEVELS RISE ABOVE 1200 PPM, THE OSA DAMPER SHALL OPEN. WHEN CO2 LEVELS DROP BELOW 800 PPM, THE OSA DAMPER SHALL CLOSE. WHERE SHOWN, AVERAGING SENSORS SHALL BE INSTALLED TO AVERAGE THE SPACE TEMPERATURE FROM MULTIPLE ZONES.

OCCUPIED HOURS TO BE DETERMINED BY THE OWNER, RECOMMENDED OCCUPIED HOURS ARE MONDAY THRU FRIDAY, 7 A.M. TO 6 P.M.

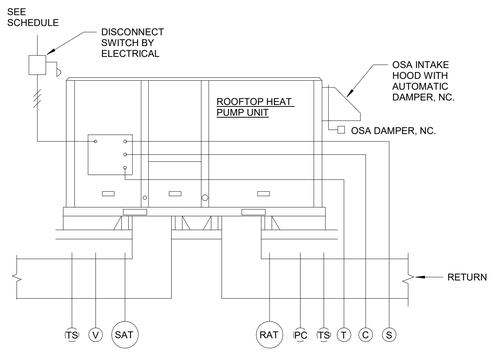
UNOCCUPIED MODE:
THE SPACE TEMPERATURE SENSOR SHALL CYCLE THE SUPPLY FAN, DX COOLING, DX HEATING, AND ELECTRIC HEAT TO MAINTAIN THE UNOCCUPIED SPACE SETPOINT (60°F HEATING / 80°F COOLING - ADJUSTABLE). THE OUTSIDE AIR DAMPER SHALL REMAIN CLOSED.

FIELD INSTALL TEMPERATURE SENSORS IN SUPPLY AND RETURN DUCTS OF ALL UNITS. SENSORS SHALL BE CAPABLE OF BEING MONITORED THRU IOT BMS/EMS.

MECHANICAL CONTRACTOR SHALL PROVIDE AND INSTALL A SINGLE TWISTED PAIR FROM THERMOSTAT TO CENTRALLY MOUNTED CONTROL PANEL.

INSTALL PARTICLE COUNTER RETURN DUCT AND VOC SENSOR IN SUPPLY DUCT (RTU-2 ONLY). COORDINATE WITH IOT.

IOT WILL SUPPLY NETWORKABLE THERMOSTAT TO BE INTEGRATED TO THE SYSTEM AND CONTROLS FOR BMS OPERATION AFTER EQUIPMENT START UP.



(T) SPACE TEMPERATURE SENSOR WITH SETPOINT ADJUSTMENT.

(C) SPACE CO2 SENSOR

(S) MONITORING SPACE TEMPERATURE SENSOR (QUANTITY AS SHOWN ON PLANS)

(SAT) FIELD INSTALLED DUCT MOUNTED SUPPLY AIR TEMPERATURE SENSOR TIED TO RTU SYSTEM AND MONITORED THRU IOT CONTROLS

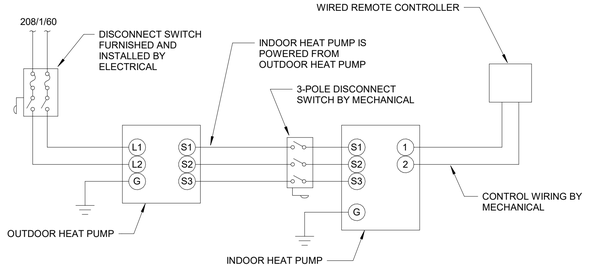
(RAT) FIELD INSTALLED DUCT MOUNTED RETURN AIR TEMPERATURE SENSOR TIED TO RTU SYSTEM AND MONITORED THRU IOT CONTROLS

2 M-503 RTU SEQUENCE AND CONTROL POINTS DETAIL
NOT TO SCALE

THE AC UNIT CONTROLLER SHALL CYCLE ON COMPRESSOR(S) TO MAINTAIN COOLING SETPOINT (74°F - ADJUSTABLE) AND HEATING SETPOINT (70°F - ADJUSTABLE).

CONTRACTOR TO PROVIDE OEM CONVENTIONAL THERMOSTAT ADAPTER WITH UNIT, PLEASE PROVIDE MITSUBISHI PAC-US445CN-1 AT PROJECT START. THE ADAPTER SHALL BE COMPATIBLE BY EITHER USING A BACNET CARD OR ALLOW INTEGRATION WITH A CONVENTIONAL THERMOSTAT.

IOT WILL SUPPLY NETWORKABLE THERMOSTAT TO BE INTEGRATED TO THE SYSTEM AND ADD CONTROLS FOR BMS OPERATION AFTER EQUIPMENT STARTUP.



3 M-503 FCU/HP SEQUENCE AND CONTROL POINTS DETAIL
NOT TO SCALE

MC TO ENGAGE A RESPRESENTATIVE FROM CARRIER (OR ALTERNATIVE MANUFACTURER) FOR START UP PRODECURES AND COORDINATION WITH IOT FOR QC PROCESS.

WORKFLOW:

1. MC TO PERFORM FULL START-UP AND CONFIGURATION OF UNIT TO FULFILL FACTORY AND JOB SCOPE REQUIREMENTS.
2. MC TO SCHEDULE AN ONBOARDING CALL WITH IOT PRIOR TO THE SITE VISIT (IOT IS OPEN FROM 8:00 AM TO 5:00 PM (CST)
   - A. PRIOR TO THE ONBOARDING CALL THE MC NEEDS TO READ THE DIGIMEP USER GUIDE
     - a. IT COVERS HOW TO USE THE APP (PG. 1-17)
       - BEST PRACTICES AND EXPECTATIONS
     - b. MEASUREMENTS TO TAKE (PG. 18)
     - c. TOOLS REQUIRED (PG. 19-20)
     - d. APP TROUBLESHOOTING (PG. 21-22)
   - B. WE WILL ALSO NEED THE FOLLOWING INFORMATION FOR THE TECHNICIAN PERFORMING THE SITE VISIT TO MAKE SURE HE/SHE HAS THE CREDENTIALS NEEDED TO LOGIN TO THE APP
     - a. FIRST AND LAST NAME
     - b. EMAIL
     - c. PHONE NUMBER
     - d. MAKE AND MODEL
3. MC TO OPEN THE DIGIMEP APP AND FOLLOW THE STEP-BY-STEP CX AND STARTUP VERIFICATION PROCESS AND SUBMIT REPORT.
4. MC AND DESIGNATED STAKEHOLDERS WILL RECEIVE A REPORT SECONDS AFTER SUBMISSION
   - A. MC WILL NEED TO REVIEW AND ADDRESS ANY DEFICIENCIES FOUND IN PERFORMANCE AND QC READINGS AND CALLOUTS.
   - B. AFTER CORRECTION OF ANY DEFICIENCIES, MC WILL OPEN THE SUBMITTED FORM, TAKE NEW READINGS TO VALIDATE UNIT IS IN PASS STATUS.
     - a. MC TO CALL IOT TO REVIEW AND VALIDATE FORM AT 512-649-5142 EXT. 7
     - b. A COMMISSIONING TECH WILL REVIEW THE FORM WITH YOU TO MAKE SURE YOU ARE GOOD TO GO.

NOTE: ALL OF THESE READINGS AND ASSET DATA, PERFORMANCE DATA AND PICTURES WILL CONFIRM QUALITY OF INSTALL AND WILL BE A PERMANENT ASSET RECORD THAT WILL VALIDATE SCOPE AND THAT THE RESULTANT PERFORMANCE IS ACCEPTED.

4 M-503 ONBOARDING START-UP REVIEW MEETING
NTS

IOT BMS NOTE:

IOT SHALL PROVIDE BMS. COORDINATE WITH IOT FOR SUPPORT PRICING. CONTACT GERRIANNE FAULHABER.

EMAIL: GERRIANNE@IOFTEAM.COM
PHONE: (224)213-6908

MECHANICAL CONTRACTOR TO SCHEDULE ON BOARDING SESSIONS WITH IOT PRIOR TO INSTALL.

STARTUP MEETING NOTE:

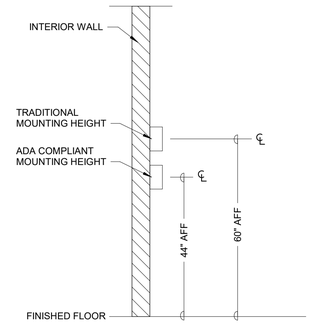
GC TO COORDINATE AN ON BOARDING EMAIL WITH THE EC, MC, CBRE PM. THE GC NEEDS TO ATTEND THE MEETING.

LOW VOLTAGE NOTE:

MC TO REVIEW IOM MANUAL FOR LOW VOLTAGE ROUTING. LOW VOLTAGE ROUTING TO BE RUN PER IOM AND NOT ON SAME CONDUIT AS HIGH VOLTAGE.

THERMOSTAT MOUNTING NOTES:

1. ALL THERMOSTATS SHALL BE INSTALLED AT 44" AFF UNLESS NOTED OTHERWISE.
2. MAINTAIN 16" FROM ADJACENT EXTERIOR WALLS. WHERE INSTALLATION IS REQUIRED ON EXTERIOR WALL, PROVIDE INSULATED MOUNTING PAD FOR BASE PLATE.
3. DISTANCE FROM CORNERS AND IF OTHER SENSORS NEARBY TO BE SIDE BY SIDE.
4. DISTANCE FROM WALL CORNERS TO BE 6" MINIMUM.
5. REFRAIN FROM INSTALLING THERMOSTAT NEAR ANY HEATING OR COOLING SOURCES; SUCH AS EQUIPMENT, WINDOWS, OR UNDER SUPPLY DIFFUSERS.
6. ENSURE THERMOSTAT IS INSTALLED IN CORRECT ORIENTATION AND ACCORDING TO ALL MANUFACTURER INSTALLATION INSTRUCTIONS.
7. COORDINATE FINAL LOCATION WITH IN FIELD CONDITIONS, MAINTAIN CLEARANCE FROM NEARBY SWITCHES AND OTHER DEVICES AS NEEDED.
8. IF OTHER SENSORS ARE LOCATED NEARBY INSTALL SIDE BY SIDE.
9. J-BOXES ARE TO BE USED ON ALL MONITORING SENSORS, THERMOSTATS AND CO2 SENSORS.



5 M-503 WALL THERMOSTAT MOUNTING DETAIL
NTS

6/30/2026 4:55:05 PM

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Rev	Date	Comments
	07/02/2026	PERMIT SET

Client:
FIFTH THIRD BANK

Project Number: 26032
Project Name:
FIFTH THIRD BANK WATSON
1,900 SF

2711 WATSON BOULEVARD
WARNER ROBBINS, GA 31093

Key Plan:

Sheet Title:
MECHANICAL CONTROLS DETAILS

Sheet Number:
M-503